

Prediction (1)

July 13th

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Supervised learning (I)

- Supervised learning is a core branch of machine learning where the goal is to learn a mapping from inputs to outputs using **labeled** data.
- You are given a dataset of **input-output pairs**:
$$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$$
- $X_i \in \mathbb{R}^p$ is a feature vector, and each Y_i (a number) is a target label or value (also called a response).
- In the tidy dataset, one of the columns is the label
- The goal is to learn a function $f: Y \approx f(X)$
- After training, we use $f(X_{n+1})$ to make predictions on **new, unseen inputs**.

Supervised learning (II)

Problem Type

Regression

Classification

Output Space

Continuous (e.g. real line)

Discrete / Categorical

Example Tasks

Predict house prices, temperatures

Predict spam/non-spam,
disease/no disease

Training data: Used to fit f .

Testing data: Used to evaluate generalization (out-of-sample performance).

Concept

Underfitting

Overfitting

Description

Model too simple; fails to capture the pattern.

Model too complex; captures noise as signal.

Goal: Find a model that balances bias and variance — the classic bias-variance trade-off.

<https://mlu-explain.github.io/bias-variance/>

Learning goal of prediction

- Given a measurement (loss function) $\mathcal{L}$, for lots of candidates in the candidate sets $\mathcal{F}$, we want to find a $f \in \mathcal{F}$, the following values can be minimized

$$\frac{1}{n} \sum_{i=1}^n \mathcal{L}(Y_i, f(X_i))$$

Problem Type	Loss Function	Formula	Description
Regression	Squared Error Loss (MSE)	$\mathcal{L}(Y_i, f(X_i)) = (Y_i - f(X_i))^2$	Common default.
	Absolute Error Loss(AES)	$\mathcal{L}(Y_i, f(X_i)) = Y_i - f(X_i) $	
Classification	0-1 Loss	$\mathcal{L}(Y_i, f(X_i)) = \mathbf{1}(Y_i \neq f(X_i))$	Counts misclassifications
	Cross-Entropy Loss	$\mathcal{L}(Y_i, f(X_i)) = -Y_i \log f(X_i)$	Measures distance between true and predicted distributions.

Take away:

- There are hundreds of thousands of prediction methods, mainly depending on the classes of

$\mathcal{L}$, and $\mathcal{F}$

- **Simple models** (e.g., linear regression) → high bias, low variance.
- **Complex models** (e.g., deep neural nets, decision trees without pruning) → low bias, high variance.
- Bias-variance trade off

Regression with MSE

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Linear class



Algorithm

Linear Regression (OLS)

<https://mlu-explain.github.io/linear-regression/>

Polynomial Regression

Ridge Regression

Lasso Regression

Elastic Net

Description

Assumes a linear relationship between input features and response.

Extends linear regression by including polynomial terms of predictors.

Adds ℓ_2 penalty to control coefficient size and reduce variance.

Adds ℓ_1 penalty to encourage sparsity in coefficients.

Combines ℓ_1 and ℓ_2 penalties.

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Nonparametric class



Algorithm

k-Nearest Neighbors Regression (k-NN)

<https://pub.towardsai.net/a-comparative-study-of-linear-and-knn-regression-a31955e6263d>

Locally Weighted Regression (LOESS/LOWESS)

Spline Regression (e.g., B-splines)

Description

Predicts based on the average of nearby neighbors.

Fits local models weighted by proximity.

Piecewise polynomials for modeling flexible curves.

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Tree-Based Methods



Algorithm

Decision Tree Regression

<https://mlu-explain.github.io/decision-tree/>

Random Forest Regression

<https://mlu-explain.github.io/random-forest/>

Gradient Boosted Regression Trees (e.g., XGBoost, LightGBM, CatBoost)

AdaBoost Regression

Description

Splits data recursively to model piecewise constant functions.

Averaging multiple trees to reduce variance (bagging is often used).

Builds trees sequentially to reduce prediction error.

Boosts weak learners (e.g., small trees) using residuals.



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Hybrid Models



Algorithm

Principal Component Regression (PCR)

Partial Least Squares (PLS)

Ensemble Regression (Stacking and Bagging)

Description

Applies PCA to features, then fits a linear model.

Projects predictors to latent space maximizing covariance with response.

Combines predictions from multiple regressors.

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Other classes including Bayes regression and deep neural networks (DNN) will be discussed later this week (by Prof. Chen)

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